

Routing and Navigation in Angular SPA

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Agenda

1. Bootstrapping router: *@angular/router*, setting the base href
2. Configuring the router
3. Router navigation using `<a routerLink="...">`
4. Programmatic navigation using *Router.navigate()*
5. Retrieving route information
6. Animating transitions for route components
7. confirming or canceling navigation with guards
8. *CanActivate*, *CanActivateChild*, *CanDeactivate*, *CanDeactivateChild* guards
9. Using *resolve* to pre-fetch data before activating a route
10. Lazy loading of feature modules

Where is The Code?

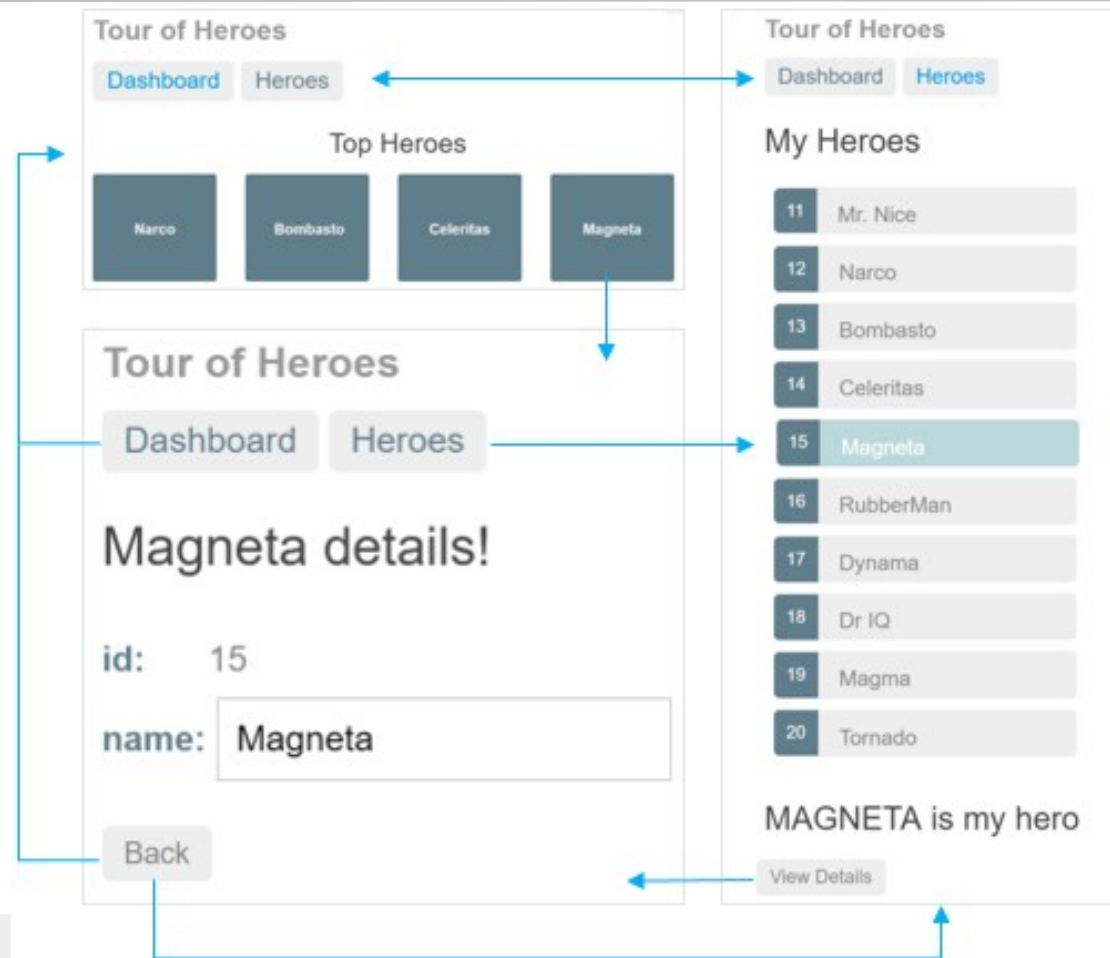
Angular 2 and TypeScript Web App Development
code is available @GitHub:

<https://github.com/iproduct/course-angular2>

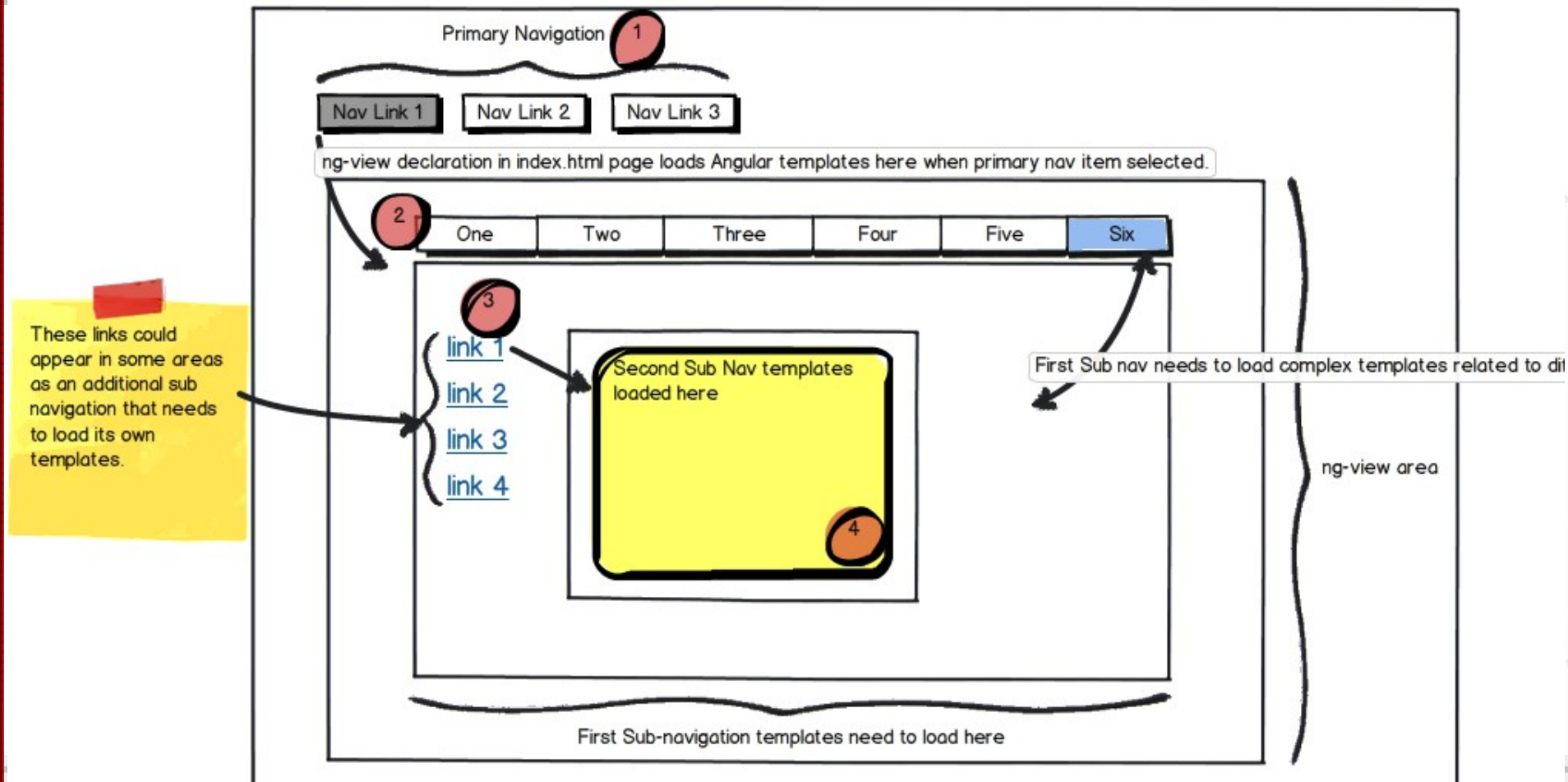
Contemporary Web Applications

- Provide better User Experience (UX) by:
 - more interactive
 - loading and reacting faster in response (or even anticipation) of user's moves
 - able to work offline
 - supporting multiple devices and screen resolutions (responsive design)
 - are following design metaphors consistently (e.g. Google Material Design - MD)
 - looking more like desktop application than static web page

Single Page Applications (SPA)



Hierarchical Routing



Source: <http://stackoverflow.com/questions/12863663/complex-nesting-of-partials-and-templates>
Author: PhillipKregg

SPA with Multiple Router Outlets

Root

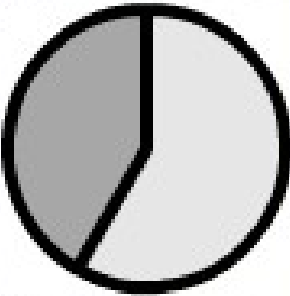
filters

Type: Age Range to Date:

tabledata

Name (job title)	Age	Nickname	Employee
Giacomo Guilizzoni Founder & CEO	34	Peldi	☒
Guido Jack Guilizzoni	4	The Guida	☐
Marco Botton Tuttofare	31		☒
Mariah MacLachlan Better Half	35	Patata	☒
Valerie Liberty COO, WOW! Division	30	Val	☒

graph



Why SPA?

- Page does not flicker – seamless (or even animated) transitions
- Less data transferred – responses are cached
- Only raw data, not markup
- Features can be loaded on demand (lazy) or in background
- Most page processing happens on the client offloading the server: REST data services + snapshots for crawlers (SEO)
- Code reuse – REST endpoints are general purpose
- Supporting multiple platforms (hybrid) → Ionic 2, NativeScript

Bootstrapping & Configuring Angular Router

- index.html: `<base href="/" />`

- app-routing-module.ts:

```
import { RouterModule } from '@angular/router';
...
@NgModule({
  imports: [ RouterModule.forRoot([
    { path: '', redirectTo: '/home', pathMatch: 'full' },
    { path: 'home', component: HomeComponent },
    { path: 'products', component: ProductListComponent },
    { path: 'users', component: UserListComponent },
    { path: 'users/:userId', component: UserDetailComponent },
    { path: '**', component: PageNotFoundComponent } ] ]),
  exports: [ RouterModule ]
})
export class AppRoutingModule {}
```

Navigation Using RouterLink Directive

- Router links:

```
<li><a routerLink="/users" routerLinkActive="active">Users</a></li>
<li><a routerLink="/admin" routerLinkActive="active">Admin</a></li>
<li><a routerLink="/login" routerLinkActive="active">Login</a></li>
```

- Using property bindings (/users/jim#qualifications?details=true):

```
<a [routerLink]="['/users/jim']" [queryParams]="{details: true}"
  fragment="qualifications">
  link to user component
</a>
```

- Using parameters array(/team/11/users/jim;details=true):

```
<a [routerLink]="['/team', teamId, 'users', userName, {details: true}]">
  link to user component
</a>
```

Router Outlets

- **app.component.html :**

```
<div class="container">
  <app-nav></app-nav>

  <!-- PRIMARY_OUTLET - here are displayed routed views by default -->
  <router-outlet></router-outlet>
  <router-outlet name='left'></router-outlet>
  <router-outlet name='right'></router-outlet>
  <router-outlet name='details' (activate)='onActivate($event)'
    (deactivate)='onDeactivate($event)'></router-outlet>
</div>
```

Handling **ActivatedRoute.params** Observable

[<https://angular.io/docs/ts/latest/guide/router.html#!#activated-route>]

```
export class ProductListComponent implements OnInit, OnDestroy {  
  constructor(private service: ProductService, private route:  
    ActivatedRoute, private router: Router, private location:Location){}  
  public ngOnInit() {  
    this.route.params  
      // (+) converts string 'id' to a number  
      .switchMap((params: Params) =>this.service.getHero(+params['id']))  
      .subscribe((hero: Hero) => this.hero = hero);  
  }  
  ...  
}
```

Programmatic Navigation using `Router.navigate()`

```
export class ProductListComponent implements OnInit, OnDestroy {  
  constructor(private service: ProductService, private route:  
    ActivatedRoute, private router: Router, private location:Location){}  
  public ngOnInit() {  
    this.route.params // highlight previously selected product  
      .forEach((params: Params) => {  
        this.selectedId = +params['selectedId'];  
      });  
  }  
  public selectItem(product: Product) {  
    this.selectedId = product.id;  
    this.router.navigate(  
      ['.', {selectedId: product.id }], {replaceUrl: true})  
      .then(isSuccess => this.router.navigate(['/product', product.id]));  
  }  
}
```

Routing with Animations I

```
import { animate, AnimationEntryMetadata, state, style, transition,
trigger } from '@angular/core';
export const slideInDownAnimation: AnimationEntryMetadata =
  trigger('routeAnimation', [
    state('*',
      style({ opacity: 1, transform: 'translateX(0)' })
    ),
    transition(':enter', [
      style({ opacity: 0, transform: 'translateX(-100%)' }),
      animate('0.6s ease-in')
    ]),
    transition(':leave', [
      animate('0.6s ease-out',
        style({ opacity: 0, transform: 'translateY(100%)' }))
    ])
  ]);
```

Routing with Animations II

```
import { Component, HostBinding } from '@angular/core';
import { slideInDownAnimation } from '../common/animations';

@Component({
  selector: 'simple-form',
  templateUrl: './home.component.html',
  animations: [ slideInDownAnimation ]
})
export class HomeComponent {
  @HostBinding('@routeAnimation') routeAnimation = true;
  @HostBinding('style.display') display = 'block';
  @HostBinding('style.width') width = '100%';
  @HostBinding('style.position') position = 'absolute';

  public imageBox = require('../../assets/img/ipt-box.png');
}
```


Routing Guards: CanDeactivate - I

```
@NgModule({  
  imports: [  
    RouterModule.forChild([  
      { path: 'products', component: ProductListComponent },  
      {  
        path: 'product/:id',  
        component: ProductDetailComponent,  
        canDeactivate: [CanDeactivateGuard],  
        data: {  
          title: 'Edit Product'  
        }  
      }  
    ])],  
  exports: [ RouterModule ]  
})  
export class ProductRoutingModule {}
```


Routing Guards: CanDeactivate - II

```
@Component({
  selector: 'product-detail',
  templateUrl: './product-detail.component.html'
})
export class ProductDetailComponent implements OnInit, OnChanges,
  CanComponentDeactivate {
  public product: Product = { id: undefined };
  constructor(private fb: FormBuilder, private route: ActivatedRoute,
    private router: Router, private location: Location, private service:
    ProductService, private dialogService: DialogService) { }
  canDeactivate(): Promise<boolean> | boolean {
    if ( shallowEquals(this.product, this.productForm.getRawValue())){
      return true;
    }
    return this.dialogService.confirm('Discard changes?');
  }
  ...
}
```

Resolving Route Data Asynchronously I

```
@NgModule({
  imports: [
    RouterModule.forChild([
      { path: 'products', component: ProductListComponent },
      {
        path: 'product/:id',
        component: ProductDetailComponent,
        canActivate: [CanDeactivateGuard],
        data: { title: 'Edit Product' },
        resolve: { product: ProductResolver }
      }
    ])
  ],
  exports: [RouterModule],
  providers: [ ProductResolver ]
})

export class ProductRoutingModule {}
```

Resolving Route Data Asynchronously II

```
import { Injectable } from '@angular/core';
import { Router, Resolve, ActivatedRouteSnapshot } from '@angular/router';
import { Product } from './product.model';
import { ProductService } from './product.service';

@Injectable()
export class ProductResolver implements Resolve<Product> {
  constructor(private service: ProductService, private router: Router){}
  public resolve(route: ActivatedRouteSnapshot): Promise<Product> {
    let id = +route.params['id'];
    return this.service.refreshProducts().then(() =>
      this.service.getProductObservable(id).take(1).toPromise());
  }
}
```

Lazy Loading of Feature Modules

```
import { Routes } from '@angular/router';
import { NotFoundComponent } from '../ui/components/not-found';
export const routes: Routes = [
  {
    path: '',
    redirectTo: '/users',
    pathMatch: 'full' },
  {
    path: 'tests',
    loadChildren: './tests/test.module#TestModule'
  },
  {
    path: 'users',
    loadChildren: './users/user.module#UserModule'
  },
  {
    path: '**',
    component: NotFoundComponent
  }
];
```

Lazy Loading – Child RoutingModule (1)

```
@NgModule({
  imports: [
    RouterModule.forChild([
      {
        path: '',
        component: UserListComponent,
        children: [{
          path: 'new',
          pathMatch: 'full',
          component: UserDetailComponent,
          data: { title: 'Add New User' }
        }, {
          path: ':id',
          component: UserDetailComponent,
          canActivate: [CanDeactivateGuard],
          data: { title: 'Edit User' },
          resolve: { user: UserResolver }
        }
      ]
    ])
  ],
```

Lazy Loading – Child RoutingModule (2)

```
...  
  
exports: [  
  RouterModule  
],  
providers: [  
  UserResolver  
]  
})  
export class UserRoutingModule { }
```



Thanks for Your Attention!

Questions?

